

JAVA PROJECT REPORT ON
COURIER PARCEL TRACKING AND DRIVING SYSTEM

Submitted by
MIRUTHU BHASHINI M
25BCS127

Under the Guidance of
Mr. Aravindh
Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **I**

DECLARATION

I hereby declare that the project entitled "**Courier parcel tracking and driving system**" submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mr Aravindh**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

____Miruthu bhashini M_____

CERTIFICATE

This is to certify that the project entitled “**COURIER PARCEL TRACKING AND DRIVING SYSTEM**” is a **bona fide work** carried out by **Miruthu Bhashini I Bsc CS B Computer Science KG College of Arts & Science** fulfillment of the requirements for the course **Java Programming**, in partial
This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10/04/26

Submitted for the Viva-Voce Examination held on ____10/04/26_____

Guide:

(Technical Trainer, KGCAS)

Head of the Department

(Technical Head, KGCAS)

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	Introduction	6
1.1	Objective	6
1.2	Overview	6
1.3	Features	7
2	Program Analysis	8
2.1	Problem Definition	8
2.2	Project Overview	8
2.3	Scopes and Limitations	9
3	System Design	10
3.1	Modules	10
4	Methodology	11
4.1	Problem Analysis	11
4.2	System Design	11
4.3	Program Development	11
4.4	Testing and Validation	12
4.5	Result and Documentation	12
4.6	Algorithm	12
4.7	Flowchart	13
5	Source Code	14-15
6	Input and Output Screens	16
7	Discussion and Conclusion	17
7.1	Discussion	17
7.2	Conclusion	17
8	References	18

ABSTRACT

The **Courier parcel tracking and driving system** is designed to enhance the efficiency, transparency, and reliability of logistics operations. This system integrates real-time parcel tracking with route optimization for delivery drivers, enabling both customers and service providers to monitor shipment progress accurately. By leveraging modern technologies such as GPS, cloud computing, and mobile applications, the system ensures timely updates and improved coordination between dispatch centers and drivers.

The parcel tracking module allows users to track their shipments from dispatch to delivery using a unique tracking ID. It provides real-time location updates, estimated delivery time, and status notifications, thereby improving customer satisfaction and trust. The system also maintains a centralized database to store parcel details, delivery history, and customer information, ensuring easy access and management of records.

The driving system focuses on optimizing delivery routes and assisting drivers in navigation. It uses intelligent algorithms to determine the shortest and most efficient paths, reducing fuel consumption and delivery time. Drivers receive route suggestions, traffic updates, and delivery schedules through a user-friendly interface, helping them manage multiple deliveries effectively and avoid delays.

Overall, the Courier Parcel Tracking and Driving System aims to streamline logistics operations by combining tracking and driving functionalities into a single platform. It improves operational efficiency, reduces manual errors, and enhances the overall delivery experience for both customers and service providers, making it a valuable solution in the modern logistics industry.

1.INTRODUCTION

The **Courier parcel tracking and driving system** is developed to improve the overall performance and efficiency of courier and logistics services. In today's fast-paced world, the demand for quick and reliable parcel delivery has increased significantly. Traditional methods of tracking and managing deliveries are often time-consuming and prone to errors, which creates the need for a more advanced and automated system.

This system provides a digital solution that allows users to track their parcels in real time. By using a unique tracking ID, customers can easily check the current location and status of their shipment at any time. This transparency helps in building trust between the service provider and the customer while also reducing the number of inquiries related to delivery status.

In addition to parcel tracking, the system also focuses on improving the efficiency of delivery drivers. It assists drivers by providing optimized routes, navigation support, and real-time traffic updates. This helps in reducing delivery time, saving fuel, and ensuring that parcels reach their destination without unnecessary delays.

The system also includes a centralized database that stores all important information such as parcel details, customer data, and delivery history. This makes it easier for administrators to manage operations, monitor performance, and generate reports when needed. It reduces manual work and improves accuracy in handling data.

Overall, the **Courier parcel tracking and driving system** plays a vital role in modernizing logistics operations. By combining tracking and driving functionalities, it enhances service quality, increases efficiency, and provides a better experience for both customers and delivery personnel.

1.1 Objective

The main objective of the **Courier parcel tracking and driving system** is to develop a simple and efficient application that automates parcel tracking and delivery management using modern technologies. The system aims to improve the accuracy, speed, and transparency of courier services by integrating real-time tracking and driver assistance features.

The specific objectives of this project are:

- To design and develop a parcel tracking system for courier services.
- To enable real-time tracking of parcels using a unique tracking ID.
- To automate the updating of parcel status and delivery information
- To reduce manual errors in tracking and delivery management.
- To provide route optimization and navigation support for delivery drivers.
- To improve delivery efficiency by reducing time and fuel consumption.

1.2 Overview

The **Courier Parcel Tracking and Driving System** is a user-friendly application developed to manage parcel tracking and delivery operations efficiently. The system provides a digital platform where users and administrators can monitor the movement of parcels in real time and manage delivery processes effectively.

The system allows users to track parcels using a unique tracking ID, which displays the current location, status, and estimated delivery time. Delivery personnel can update parcel status at different stages such as dispatch, in transit, and delivered. This ensures accurate and up-to-date information is available to both customers and service providers.

The system also supports delivery drivers by providing route optimization and navigation assistance. It helps drivers choose the shortest and most efficient routes, reducing delivery time and fuel consumption. The application can be developed as a **web-based system or a mobile application** for better accessibility and convenience.

This project demonstrates the practical use of programming and modern technologies in developing real-world logistics solutions. It is especially useful for courier companies and delivery services to improve efficiency, reduce errors, and enhance customer satisfaction.

1.3 Features

The **Courier ParcelTrackingandDrivingSystem** includes the following main features:

- **Parcel Tracking** – Allows users to track parcels in real time using a unique tracking ID.
- **Status Updates** – Enables updating of parcel status such as dispatched, in transit, out for delivery, and delivered.
- **Real-Time Location** – Provides live location tracking of parcels using GPS technology.
- **Route Optimization** – Suggests the shortest and most efficient routes for delivery drivers.
- **Driver Assistance** – Helps drivers with navigation and delivery scheduling.
- **Centralized Database** – Stores parcel details, customer information, and delivery history in an organized manner.
- **Notification System** – Sends updates and alerts to customers regarding parcel status and delivery.
- **User-Friendly Interface** – Offers a simple and easy-to-use interface for both customers and administrators.

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the system requirements are carefully studied and analyzed. This stage helps in understanding how the Courier Parcel Tracking and Driving System should function and what features are needed to ensure efficient parcel delivery and tracking. It focuses on identifying existing issues in traditional courier systems and proposing a modern, automated solution.

2.1 Problem Definition

In many courier services, parcel tracking and delivery management are often handled manually or with limited digital support. This leads to several challenges.

Some common problems include:

- Lack of real-time parcel tracking
- Delays in delivery updates
- Difficulty in managing multiple deliveries
- Poor route planning for drivers
- Inaccurate or missing delivery records

These issues increase the workload for courier staff and may result in delayed deliveries and customer dissatisfaction.

Therefore, there is a need for an automated system that can provide real-time tracking, efficient route management, and accurate delivery updates. The Courier Parcel Tracking and Driving System is designed to overcome these problems by offering a reliable and computerized solution.

2.2 Project Overview

The Courier Parcel Tracking and Driving System is developed to simplify parcel tracking and delivery operations. The system allows users to track parcels using a unique tracking ID and provides real-time updates on parcel status and location.

The system includes the following functionalities:

- Tracking parcels using a unique ID
- Updating parcel status at different stages
- Providing real-time location details
- Assisting drivers with route optimization
- Maintaining delivery records in a centralized database

This project demonstrates the use of programming concepts and modern technologies. It can be implemented as a web-based or mobile application for better accessibility and usability.

2.3 Scope and Limitations

Scope

The scope of the Courier Parcel Tracking and Driving System includes the following:

- Managing parcel information and tracking details
- Providing real-time tracking services to users
- Assisting delivery drivers with navigation and routes
- Storing and managing delivery data efficiently
- Improving overall courier service performance

The system can be used in:

- Courier companies
- Delivery service providers
- E-commerce logistics
- Postal services
- Small and large shipping businesses

Limitations

- Requires internet connectivity for real-time tracking
- GPS accuracy may vary in certain locations
- Initial setup cost for system implementation
- Limited functionality if not integrated with advanced technologies
- Depends on timely updates from delivery personnel

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the Courier Parcel Tracking and Driving System is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Courier Parcel Tracking and Driving System is designed as an integrated application where users can track parcels using a tracking ID and delivery staff can manage parcel delivery effectively. The system ensures smooth coordination between parcel tracking and driver management.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Courier Parcel Tracking and Driving System are explained below:

1.Parcel Registration Module

This module is responsible for entering parcel details into the system. It includes sender details, receiver details, parcel description, and delivery address.

Each parcel is assigned a unique tracking ID.

Example:

Tracking ID: CP1001

Sender: Ram

Receiver: Sita

Location: Chennai

2.Parcel Tracking Module

This module allows users to track the parcel using the tracking ID.

Functions of this module include:

- Displaying current parcel status
- Showing parcel location
- Providing delivery updates

3.Status Update Module

This module is used by delivery staff to update parcel status at different stages.

Functions include:

- Updating status (Dispatched, In Transit, Delivered)
- Maintaining accurate tracking information
- Ensuring real-time update

4.Notification Module

This module sends notifications to customers regarding parcel updates.

Examples:

- Parcel dispatched
- Out for delivery
- Delivered successfully

5.Exit Module

This module allows the user to exit the system safely.

When selected, the application terminates properly.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Courier Parcel Tracking System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of courier service operations.

The main problems identified in manual tracking systems are:

- Difficulty in tracking parcel status manually.
- Delay in updating parcel information.
- Lack of real-time tracking for customers.
- Errors in maintaining parcel records.
- Difficulty in handling multiple parcels efficiently.

Based on these problems, the system requirements were defined to develop an automated parcel tracking system that improves efficiency and provides accurate tracking information.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Parcel Booking Module
- Tracking Module
- Status Update Module
- Delivery Management Module
- Report Generation Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system and track parcels efficiently.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of menu-driven logic
- Writing code for parcel tracking and status updates
- Handling user input (tracking ID, parcel details)
- Displaying parcel status and delivery information

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter valid tracking ID	Parcel status displayed correctly
Enter invalid tracking ID	Error message displayed
Book new parcel	Tracking ID generated successfully
Update parcel status	Status updated correctly
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system allows users to track parcels and displays the current delivery status based on the tracking ID.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

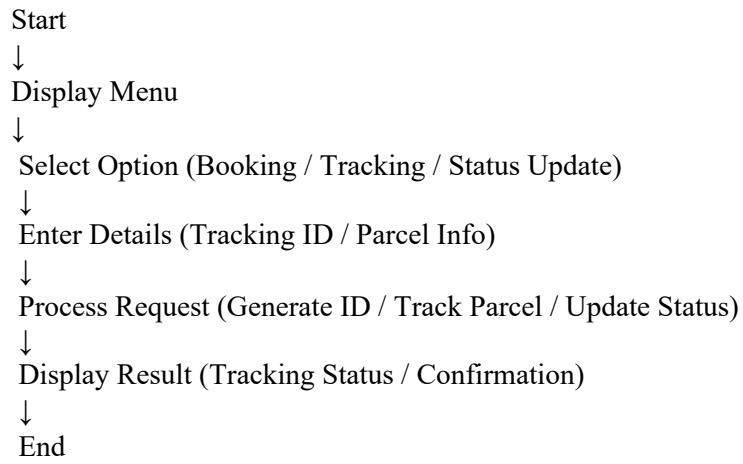
Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Courier Parcel Tracking System:

1. Start
2. Initialize system and load courier data
3. Display main menu to user
4. Accept user input (track, update, assign, report)
6. If track, enter tracking number and show status
- If update, enter new status and update database
7. If assign, assign driver to courier
8. If report, generate and display report
9. Validate all inputs for errors
10. Display results or error messages
11. End

4.7. Flowchart



5.SOURCE CODE

```
// Main class
public class CourierTrackingSystem {
public static void main(String[] args) {
System.out.println("Courier Parcel Tracking System");

} } // Menu logic

import java.util.Scanner;

Scanner sc = new Scanner(System.in);
int choice;
System.out.println("1. Book Parcel");
System.out.println("2. Track Parcel");
System.out.println("3. Update Status");
System.out.println("4. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch(choice) {
case 1:
System.out.println("Parcel Booking");
break;
case 2:
System.out.println("Parcel Tracking");
break;
case 3:
System.out.println("Status Update");
break;
case 4:
System.out.println("Exit");
break;
default:
System.out.println("Invalid Choice");
}

// parcel processing logic

int trackingId;
String status = "Shipped";
System.out.print("Enter Tracking ID: ");
trackingId = sc.nextInt();
System.out.println("Current Status: " + status);
// Example Calculation (Same like billing format)
int baseCharge = 100;
int weight = 3;
int total = baseCharge * weight;
int gst = (int)(total * 0.05);
int finalAmount = total + gst;
```

```
System.out.println("Total Charge: " + total);  
System.out.println("GST: " + gst);  
System.out.println("Final Amount: " + finalAmount);
```

6.INPUT AND OUTPUT SCREENS

Fig1 : Program Menu

```
Courier System Menu:  
1. Calculate Billing  
2. Exit
```

Fig2 : Item selection

```
Enter price: 500  
Enter quantity: 2
```

Fig3 : Billing output

```
Total: 1000.0  
GST (5%): 50.0  
Final Bill: 1050.0
```

Fig4 : Final Invoice

```
Courier System Menu:  
1. Calculate Billing  
2. Exit  
1  
Enter price: 500  
Enter quantity: 2  
Total: 1000.0  
GST (5%): 50.0  
Final Bill: 1050.0
```


7.DISCUSSION & CONCLUSION

7.1 Discussion

TheCourierParcelTracking System was developed to automate the process of tracking parcels in courier services. The system allows users to book parcels, enter details, and track the status of shipments using a tracking ID.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully manages parcel information and displays accurate tracking status.

The system was tested with different inputs to ensure that it performs all operations correctly. The results show that the system reduces manual effort and improves the efficiency of parcel tracking.

7.2 Conclusion

TheCourierParcelTracking System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world application.

The system automates parcel tracking, reduces errors in maintaining records, and provides quick access to parcel status. It simplifies courier operations by allowing easy tracking and management of parcels.

This project helps students understand how Java programming can be used to develop practical systems. The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, and real-time tracking using web or mobile applications.

8.REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*

Websites

- <https://www.oracle.com/java>
- <https://www.geeksforgeeks.org>